



Market Briefs Pro

Read less news

Good morning, Pro Briefers!

You may have a friend who tinkers around with a 3D printer at home, but researchers at Harvard recently managed to print functioning human blood vessels.

It sounds like science fiction, but it's not - it's what we've identified as an Innovation Shift: Advancements in 3D printing are revolutionizing the healthcare industry.

How? Companies that create medical-grade 3D printers and software, or those that utilize the technology are seeing a boom.

This week, we're going to show how to invest in this unique opportunity - from the machines fueling these innovations, all the way through hospitals getting them into the hands of patients.

Now, the science behind both 3D printing technologies and biomedical engineering is very complex, but we're going to keep things simple.

There are two main things investors need to know:

- 3D printing in medicine is rapidly evolving.
- Companies are using it to generate profits.

Global Healthcare 3D Printing Market Forecast (2024–2032)



Market Statistics

Market Value (2023)
\$1.2B

Market Value (2032)
\$13.8B

CAGR (2024-2032)
18.6%



Regional Statistics

North America Market Size (2023)
\$1.1B



Segment Statistics

Syringe-Based Segment
Market Size (2023): **\$1.2B**

Fused Deposition Modeling (FDM) Segment
Market Share (2023): **42.5%**

Medical Segment
Market Size (2023): **\$6.8B**

Polymers Segment
CAGR (2024-2032): **19%**



Market Briefs **Pro**

Data *via* GMI

- The global healthcare 3D printing market was valued at \$2.9 billion in 2023.
- It's on track to reach \$13.8 billion by 2032, a CAGR of 18.6%

Definition: CAGR is Compound Annual Growth Rate - the compounded rate at which an investment grows over time.

As a whole, the 3D printing industry has struggled, but there's plenty of printing tech that is already in hospitals and helping patients.

Our analysts scored an exclusive interview with Allison Harbaugh, a medical business manager for Stratasys, one of the most prominent manufacturers of 3D printers.

She painted a vivid picture of how this technology is helping patients:

"Imagine your surgeon presents you with a 3D printed model of your exact liver and demonstrates not only what is wrong, but the approach they plan to take in removal. The level of patient understanding and confidence is greatly impacted."

And some of these options may already be available in a hospital near you.

U.S. healthcare has a history of strong profit margins

- In 2024 the medical equipment and supplies industry reported a gross profit margin of over 53%

Where equipment and supplies intersect with the latest 3D printing technology is where we've seen this shift come to life.

We want your feedback: Tell us what you think about this edition of Market Briefs Pro [by clicking here](#).

TECHNOLOGY

3D Printing Companies

You can't have 3D printing in healthcare without...*well, 3D printers*.

These printers also need dedicated software to power them.

But very few 3d printing companies meet the strict criteria for medical use.

Fact: Most printers aren't designed for medical use. Instead these printers are used to print metal or non-medical plastics.

Medical printers need to:

- Use medical-grade materials
- Print very specific and complex designs
- Use HIPAA-compliant software

Definition: HIPAA is the Health Insurance Portability and Accountability Act that protects sensitive patient information.

The Future Of 3D Printing In Medicine: Applications & Innovations

Medical Products & Equipment

Quickly Create Devices In
Response To Patient Demand

Anatomical Models For Surgical Planning & Education

Significant Reductions In
Operative Time & Cost

Tissues & Biomaterials

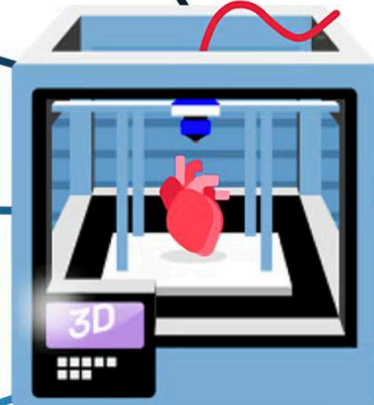
Faster & More Precise Than Traditional
Methods Of Building Organs By Hand

Prosthetics & Implants

Fast & Affordable

Personalized Drugs

More Effective Treatment



Market Briefs Pro

Data via TMF

This has left medical printing to be a small, but very lucrative niche.

Stratasys (SYSS), the company that spoke to our analysts, is largely considered to be the leader in medical printer technology.

When asked about the company's goals in medical, Allison said:

"To broaden the adoption of patient-specific 3D printed models for treatment, diagnosis, and patient understanding...3D printing for surgical planning [should] be a standard for care, not just for extremely complex cases."

Why is Stratasys so successful? Its success in the industry, and specifically in the medical field, stems from a type of printer known as "PolyJet."

What's that? PolyJet printers are capable of using multiple materials as well as printing extremely detailed textures.



Statasys claims that with their PolyJet technology, you can "produce realistic models that mimic porous bone and soft tissue," helping surgeons in things like simulation and training.

Leveraging this technology, Statasys received a globally recognized certification for medical device manufacturing in February of 2025.

This certification has strict requirements and takes years to obtain, and Statasys is one of the only 3D printer manufacturers that has announced receiving it.

- Company shares are up more than 30% since the announcement.
- Statasys has also beaten Wall Street's projections since obtaining the certification.



Stratasys shares have struggled alongside the 3D printing industry over the last five years, but have gained momentum with developments in healthcare tech - rising nearly 64% in the past six months.

While Stratasys seems to be ahead of the pack, it isn't the only 3D printer manufacturer capitalizing on medical printing.

EOS has been leaning into medical to try and increase revenue, partnering with hospital companies to create a point-of-care center that specializes in 3D printed orthopedics.

In late 2024, EOS also debuted a new printer with a major focus on its medical capabilities.

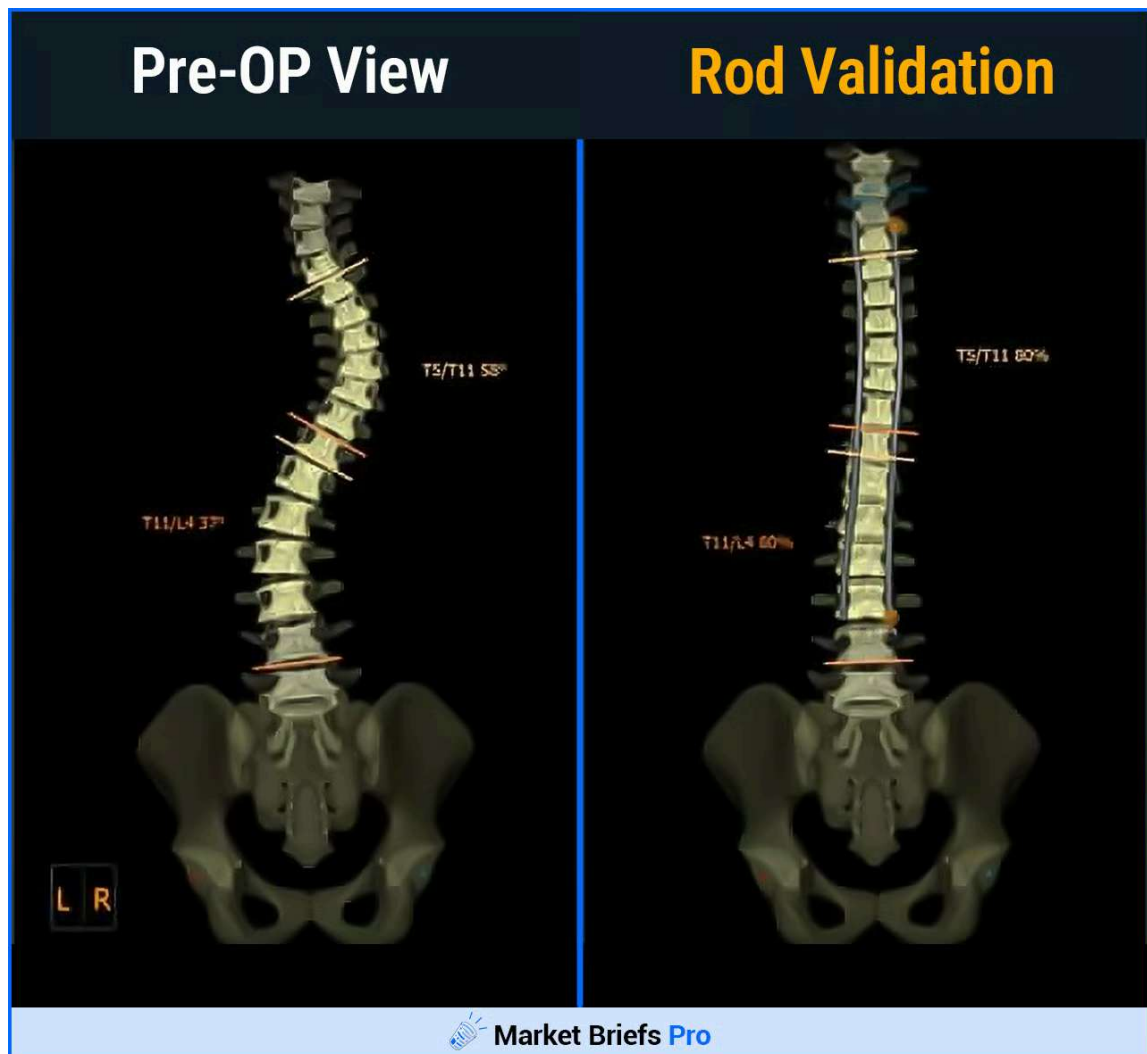


Image [via](#) Rasouli Spine

- The scanning technology from an EOS printer can be used to help map out patient orthopedics.

EOS is a privately held company, but it shows that established printing manufacturers are moving into the healthcare industry for its profit potential.

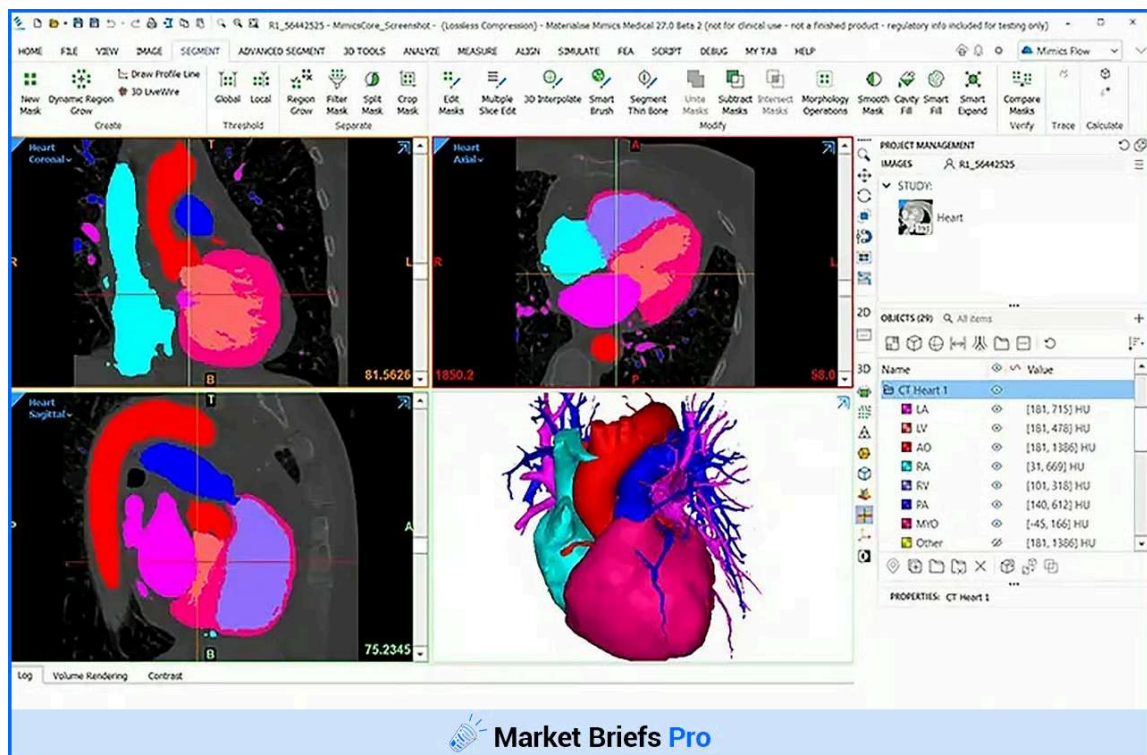
The other major component of medical printing is the software that's used to manage printers and their data.

Due to HIPAA, the healthcare industry has some of the strictest requirements when it comes to data and visibility, and much of the software that manages these printers is non-compliant.

One of the 3D printing software companies that has stood out for its medical compliance is Materialise (MTLS).

Materialise has taken major steps toward making its products HIPAA-compliant.

The result: As Materialise leans into healthcare, the once-struggling company has managed to not only stabilize but also grow.



Materialise 3D printing software connects a model of a heart to a medical printer
Image via Materialise

Since 2023, Materialise has made enormous strides in healthcare.

- In 2023 the company announced its software would power a medical implant printing facility.
- In May of 2024, Materialise software was used in the world's first successful whole-eye transplant.
- In July of 2024, the company announced it had acquired a Belgian AI company to expand its heart disease treatment.

Materialise was also the first 3D printing software company to acquire FDA clearance for use in medical diagnostics.

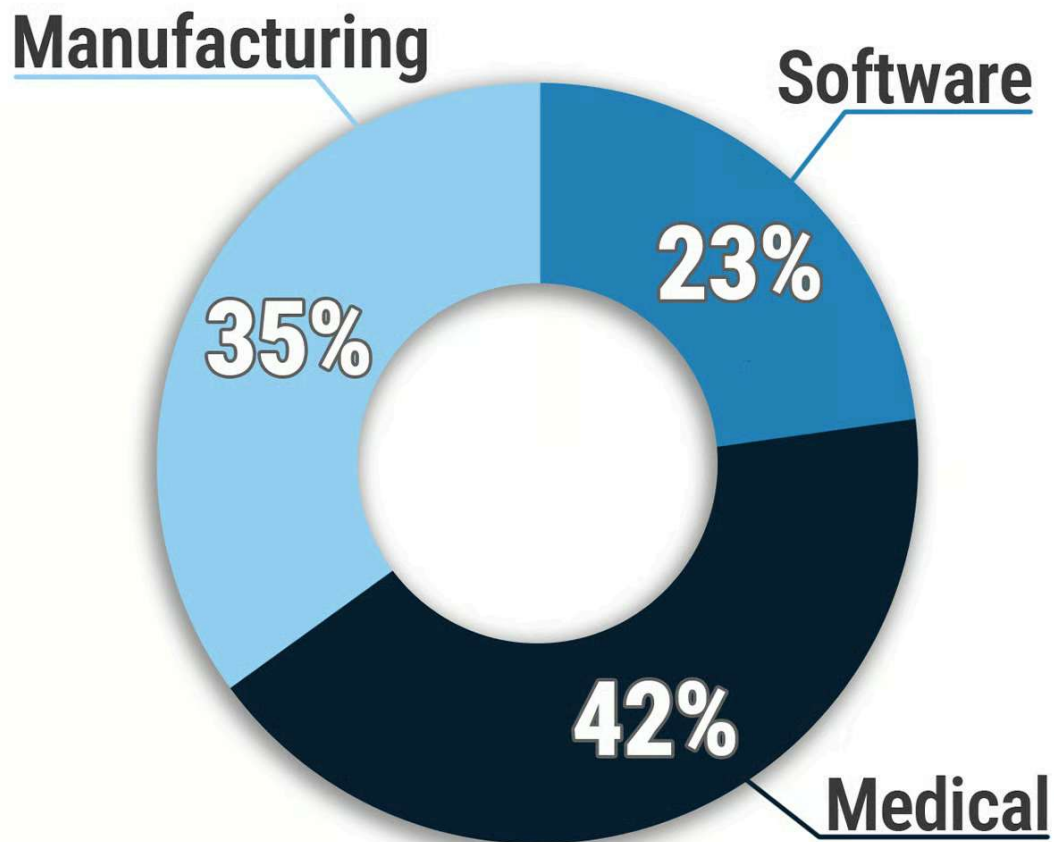


Data [via](#) Yahoo! Finance

Until recently, Materialise had been on a slow and steady decline alongside most of the 3D printing industry, dropping in both share price and revenue.

But it is amid these medical innovations that the company began to turn around.

Materialise Q3 2020 Revenue Breakdown By Segment



 Market Briefs Pro

Data [via](#) Materialise

- Medical accounts for more than 40% of Materialise's revenue.

Investors looking for a broader approach to this industry can find these companies in the iShares U.S. Medical Devices ETF (IHI), which features many companies producing advanced equipment like 3D printers.

iShares U.S. Medical Devices ETF (IHI)

5 Year Price

64.35
(+41.44%)



Market Briefs Pro

Data via Yahoo! Finance

APPLICATIONS

Medical Device Companies

Printers and software may meet the technical requirements of the healthcare industry, but there's still one more key component - *they have to actually make something medical.*

The true innovation is being found in companies using these printers to produce new devices and patient care techniques.

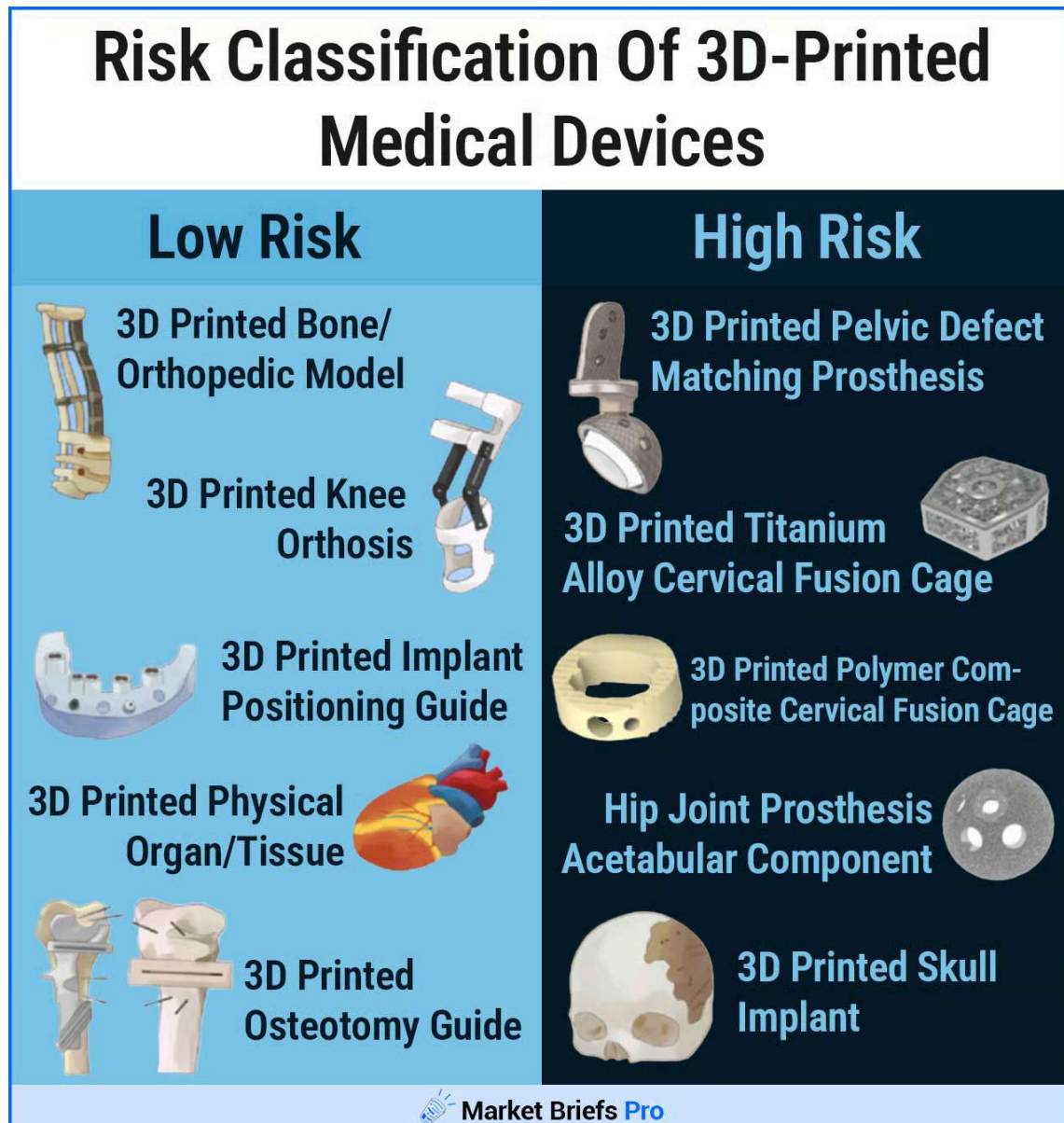
Our analysts also spoke to Dr. Oleg Krivichkin, a dentist using 3D printers in his practice.

He described how the machines have revolutionized his practice:

"The patient comes in with very bad teeth...what we can do with printers now, we take his teeth and same-day print and implant to use as a temporary."

Dr. Krivichkin said that common dental implants can be printed in as little as 8 minutes, and at a greatly reduced cost.

And it's not just dental - remember those Harvard researchers? There have also been significant strides in tissue and organ printing.



Data [via](#) ResearchGate

One of the companies that has a long history of innovations in medical printing is Organovo Holdings (ONVO), a bioprinting company that got its start by printing human tissue.

Saying that Organovo has struggled is a bit of an understatement - the company is down 97% since its 2012 debut and currently trades at under

\$1 a share - the result of over a decade with no commercially viable product.



Data [via](#) Yahoo! Finance

Down, but not out: Recently, the company has pivoted away from tissue printing and is now focused on using printers in the development of new drugs.

The company hopes it can use printers to create completely new treatments for many common diseases, finally giving it a commercial product.

With new and promising drug treatments currently before the FDA, some experts believe that Organovo is on the cusp of a comeback.

But the company hasn't totally abandoned its tissue printing roots - it's also looking to print kidney tissue to be sold for testing.

Say what? Many drug treatments can harm human kidneys, and unfortunately, there aren't a lot of ways to test drugs and dosages without...

human kidneys.

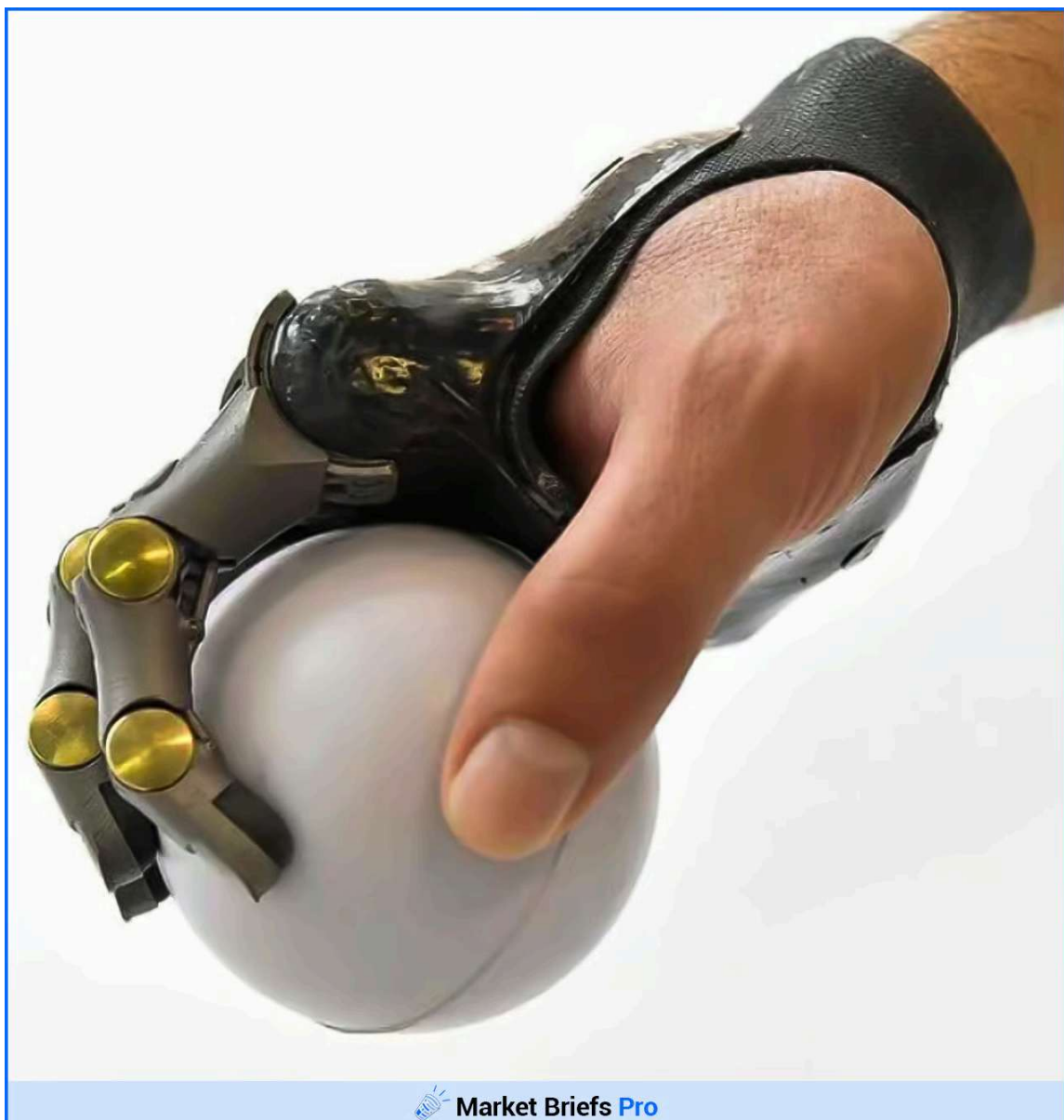
By printing kidney tissue, Organovo can not only test its drugs but potentially sell it to other pharmaceutical companies for their own testing.

If it can successfully print kidney tissue, the company will have something to sell in the interim while its experimental drugs seek FDA approval.

However, the company is on the clock - shares are currently dipping below the Nasdaq's minimum price and it could potentially be delisted from the exchange.

Soon, Organovo will either become delisted from the market or see massive returns.

There are also plenty of options for investors who don't want to take the risk of a penny stock - 3D Systems Corporation (DDD) is a pioneer in medical printing that has seen recent success.



- 3D Systems has made a name for itself with advanced prosthetics, such as partial-hand replacements (pictured above).

3D Systems is at the forefront of implant and prosthetic printing, which has been one of the most successful applications of the technology.

While it has material restrictions, printing allows for fairly complex designs and internal structures, which has led to the development of more capable prosthetic limbs.

Partnering with hospitals, 3D Systems has created what's called point-of-care printing facilities.

What's that? Instead of printing prosthetics in a factory and shipping them out, patients can get their bodies scanned and have custom-made prosthetics printed on-site.

Ease of access and major partnerships are some of the reasons why 3D Systems has seen recent success.



The company has been on a somewhat steady decline in share price since a boom in 2021, down around 60% in the last five years, but is up 100% in the last 6 months.

Innovations and increased patient access have investors taking an interest.

While the company's EPS was negative through Q3 of 2024, the final report for last year could be a make-or-break moment for the stock.

There aren't ETFs that specialize in this particular niche, with many of these companies being found in either general 3D printing or biotech ETFs.

USERS

Medical Groups

The hospitals that partner with companies like 3D Systems are the last major piece to this medical printing puzzle.

Those point-of-care facilities? They're attached to hospitals.



 **Market Briefs Pro**

*A 3D printing lab in a medical facility
Image [via](#) Materialise*

The companies we've covered so far make machines or products for patient care, but there are strict regulations surrounding patient care itself.

Medical groups and hospitals have the doctors and infrastructure to get this technology and its benefits into the hands of patients.

One of 3D Systems' biggest partners is also one of the biggest names in patient care - United Therapeutics Corporation (UTHR).

United Therapeutics began its 3D printing journey by developing 3D-printed lung scaffolds, which mimic the complex structure of a lung and help lung tissue grow.

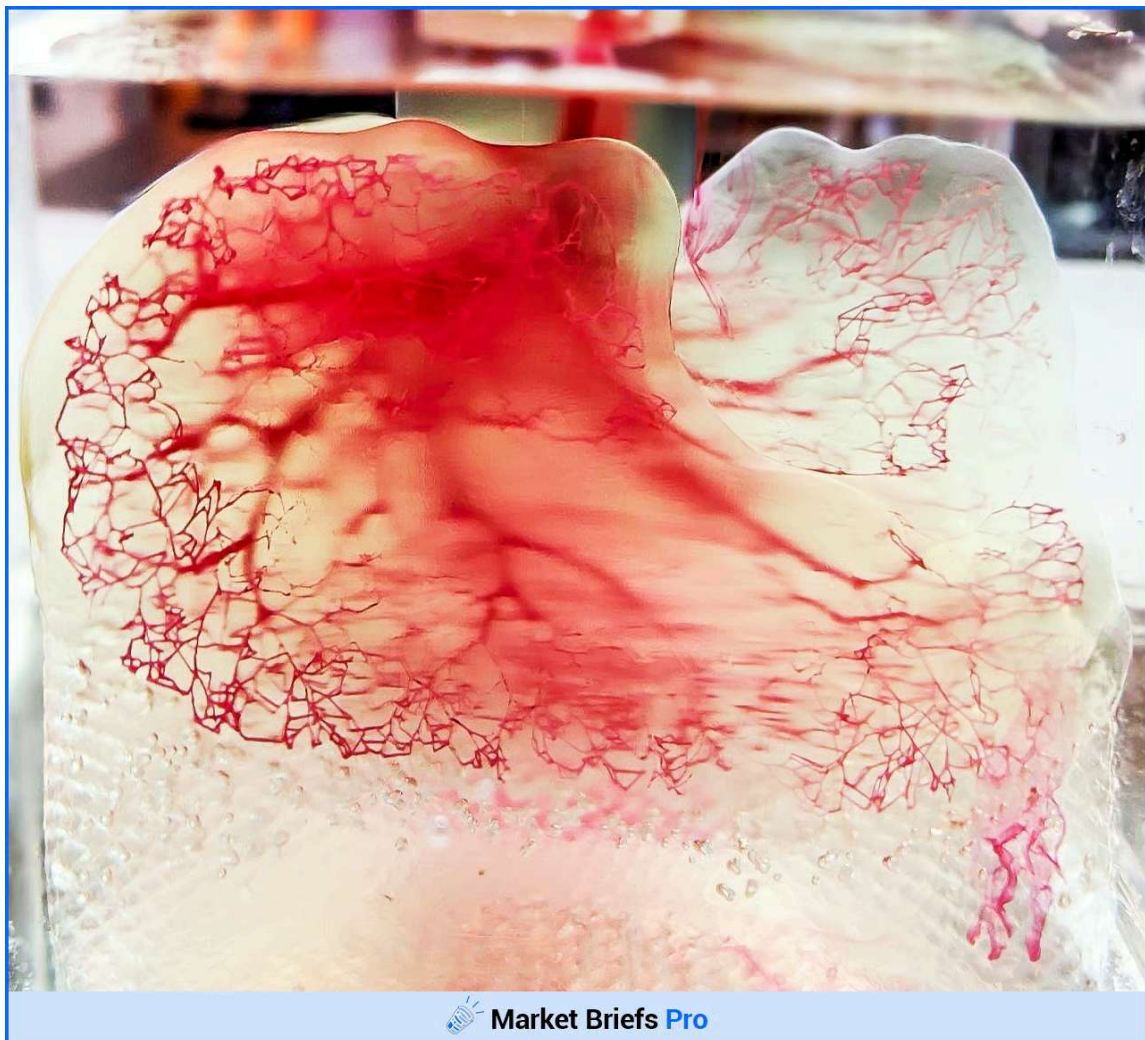


Image [via](#) Industry of Things

- Lung tissue grows around the structure provided by a 3D-printed scaffold.

It's considered a major step in the creation of printable lungs for transplants. These lungs would also be modeled after a patient's existing lungs and using their DNA, greatly reducing the chances of rejection.

United Therapeutics runs five major medical campuses across the U.S. that specialize in advanced care like this.

These advancements in tech have also advanced company revenues, which grew 20% in 2023 to \$2.3 billion, and saw additional growth of 23% YOY in its latest report (Q3 2024).

United Therapeutics Corporation (UTHR)

5 Year Price →

323.62
(+221.08%)



 **Market Briefs Pro**

Data [via](#) Yahoo! Finance

- United Therapeutics shares have seen strong and stable growth, up 255% in the last five years and 72% in the last year.

While United Therapeutics is the largest medical group utilizing this technology, it isn't the only one.

Many of the hospital groups that specialize in 3D printing or have advanced 3D printing labs are private or government-owned institutions.

While investors are unable to acquire shares of facilities like the Austin Hospital in Melbourne, Australia, or the Policlinico San Matteo in Pavia, Italy, these hospitals contain some of the most advanced labs in the world.

Those seeking to track this shift would be wise to keep these hospitals on their radar, as potential partnerships with other companies in this report could be a major boom.

REVIEW

Possible Risks

The field of medical printing has made significant strides, but that doesn't mean it is without risk.

The FDA and its European counterpart (the EMA) have extremely strict regulations around tissue, transplants, and other products from bioprinting.

As we mentioned earlier, HIPAA privacy requirements mean many of the most cutting-edge machines and software simply avoid medical printing altogether.

The combination of high capital costs and long R&D cycles also means many companies struggle to reach profitability.

Example: While Organovo's pivot may lead it to profitability, the company has been in operation for over a decade without a commercially viable product.

This technology is also rapidly evolving, meaning that it's difficult for regulators to keep up, making it even more difficult to gain approval for new products to reach patients.

Printing The Future

Despite these challenges, several companies have been able to find great success in this niche.

The 3D printing industry has been an untamed frontier for a long time, with a lot of companies breaking onto the scene and failing.

But lately, we've seen some long-standing winners emerge, and their ability to work in healthcare is a large part of that.

This tech has also reinforced dominance for some of the oldest (and largest) medical groups in the world.

Fact: This technology can be used to benefit patients - even if the exact methods haven't been figured out yet.

Those using the technology are optimistic as well, with Dr. Krivichkin saying he believes "everything will be printed" in the coming years.

As companies and hospitals crack the code on patient care, they could also print up some major profits for investors.



- Briefs Media Team

Subscribe to Market Briefs [here](#).

*The Briefs Media team often invests in the same stocks and assets we cover.

Why? Because we believe in practicing what we preach (and preaching what we practice).

But keep in mind: We're not financial advisors. Everything we provide is for education. You've got to do your own due diligence and think about what makes sense for your own financial situation. Investing has risks.

You are never guaranteed to make money when you invest, you might even lose money. If you're looking for personalized financial advice, we highly recommend speaking with a licensed financial advisor.

Our mission? It's simple - to help you be better with money.

Sent to: hello@briefs.co

[Unsubscribe](#)

Briefs Media, LLC, 100 Renaissance Ctr, Ste 1014 #43212, Detroit, Michigan 48243, United States